

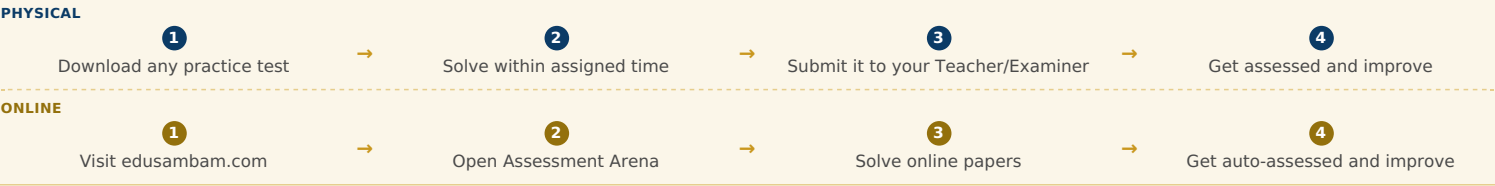
EDUSAMBAM ASSESSMENT SERIES

Physics Comprehensive Practice Test 2

PRINTABLE PRACTICE WORKSHEET

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YOUR PHYSICAL & ONLINE SELF-ASSESSMENT ROADMAP



Candidate Name:		Roll No.:	
Class / Grade:		Date:	
School / Institution:		Test Centre:	

Subject	Physics	Topic	Physics Comprehensive Practice Test 2	Total Marks	100
Total Questions	45	Time Allowed	1 Hour (60 minutes)	Version	A

GENERAL INSTRUCTIONS

- 1. Read each question carefully before answering.
- 2. For Section A and Section B, mark one answer only.
- 3. For Sections C and E, write your answer clearly in the space provided.
- 4. For Section D, write the correct letter from Column B against each item in Column A.
- 5. Do not use correction fluid. Keep all answers clear and legible.
- 6. This worksheet is prepared from the EDUSAMBAM practice assessment supplied for this paper.

PAPER PATTERN

Section	Questions	Marks Each	Total
Section A — Multiple Choice	15	2	30
Section B — True / False	10	2	20
Section C — Fill in the Blank	10	2	20
Section D — Matching	5 pairs	2	10
Section E — Short Answer	5	4	20
Total	45	—	100

Section A — Multiple Choice30 marks

1. What is the standard unit used to measure energy?

A. Newton

B. Watt

C. Joule

D. Kelvin

Answer: ☐A ☐B ☐C ☐D

Basic • 2 pts

2. What is Ohm's Law?

A. Force = Mass × Acceleration

B. Voltage = Current × Resistance

C. Work = Force × Distance

D. Energy = Mass × Speed

Answer: ☐A ☐B ☐C ☐D

Basic • 2 pts

3. According to Newton's First Law, what happens to an object in motion if no unbalanced force acts on it?

A. It gradually speeds up on its own

B. It stops immediately

C. It keeps moving at a constant velocity

D. It changes direction randomly

Answer: ☐A ☐B ☐C ☐D

Basic • 2 pts

4. According to Newton's Law of Universal Gravitation, what happens to gravitational pull as distance increases?

A. It stays exactly the same

B. It increases proportionally

C. It weakens rapidly, following an inverse square law

D. It reverses direction

Answer: ☐A ☐B ☐C ☐D

Basic • 2 pts

5. What is heat?

A. Another word for temperature

B. A measurement of mass

C. The transfer of thermal energy from a hotter object to a cooler one

D. A type of force

Answer: ☐A ☐B ☐C ☐D

Intermediate • 2 pts

6. What does the Law of Reflection state?

A. Light always bends when it reflects

B. Reflected light always travels slower

C. The angle of incidence always equals the angle of reflection

D. Light never reflects off rough surfaces

Answer: ☐A ☐B ☐C ☐D

7. In which state are particles tightly packed in fixed positions, only vibrating in place?

A. Gas

B. Plasma

C. Solid

D. Liquid

Answer: ☐A ☐B ☐C ☐D

8. If you run a full lap of a 400m track and end up exactly where you started, what is your displacement?

A. 400m

B. 200m

C. 0m

D. It cannot be calculated

Answer: ☐A ☐B ☐C ☐D

9. Why does a sharp knife cut more easily than a blunt one, with the same push force?

A. Sharp knives weigh less

B. Sharp knives are always made of different metal

C. A sharp edge concentrates force onto a smaller area, producing greater pressure

D. Sharp knives generate their own buoyant force

Answer: ☐A ☐B ☐C ☐D

10. Why can't sound travel through the vacuum of space?

A. Space is too cold for sound

B. There are no particles present for the sound wave to travel through

C. Sound moves too fast for space

D. Gravity blocks sound waves

Answer: ☐A ☐B ☐C ☐D

11. What makes physics different from a science like biology, according to the article?

A. Physics doesn't use experiments

B. Physics only studies living things

C. Physics explains "why" using precise, testable, mathematical laws

D. Physics has no connection to other sciences

Answer: ☐A ☐B ☐C ☐D

12. What is the formula for work?

A. Work = Force ÷ Distance

B. Work = Force × Distance

C. Work = Mass × Acceleration

D. Work = Distance ÷ Time

Answer: ☐A ☐B ☐C ☐D

13. What energy transformation happens in a solar panel?

A. Electrical → Light

B. Sound → Electrical

C. Chemical → Kinetic

D. Light → Electrical

Answer: ☐A ☐B ☐C ☐D

14. Why do sunsets often look red or orange?

A. The Sun itself changes color at sunset

B. Sunlight passes through more atmosphere, scattering away most of the blue light first

C. There is no scientific explanation for this

D. Red light is absorbed completely at sunset

Answer: ☐A ☐B ☐C ☐D

15. What did James Watt actually contribute, historically?

A. Major improvements to the steam engine's efficiency

B. He invented the very first engine of any kind

C. He discovered electricity

D. He invented the wheel

Answer: ☐A ☐B ☐C ☐D

Section B — True / False

20 marks

16. Chemical energy is stored in the bonds between atoms and is found in food and fuels.

A. True ☐

B. False ☐

17. Rubber and plastic are commonly used as electrical conductors.

A. True ☐

B. False ☐

18. Like magnetic poles repel and unlike poles attract.

A. True ☐

B. False ☐

19. Newton’s Third Law says action and reaction forces act on the same object.

A. True ☐

B. False ☐

20. Weight can change when local gravitational strength changes, while mass remains constant.

A. True ☐

B. False ☐

21. Heat always flows naturally from a colder object to a hotter object.
A. True ☐ B. False ☐

22. A good conductor allows electric charge to move relatively easily.
A. True ☐ B. False ☐

23. Pitch mainly depends on the frequency of a sound wave.
A. True ☐ B. False ☐

24. Ultrasound has a frequency below 20 Hz.
A. True ☐ B. False ☐

25. In an ideal energy transformation, total energy is conserved even when its form changes.
A. True ☐ B. False ☐

Section C — Fill in the Blank

20 marks

26. Ohm’s Law is $V = I \times \rule{1cm}{0.4pt}$.
Answer:

27. The two poles of a magnet are called north and .
Answer:

28. The force needed to accelerate a 2 kg object at 3 m/s² is N.
Answer:

29. Gravitational force follows an inverse law with distance.
Answer:

30. The SI unit of temperature used in the article’s temperature-scale discussion is .
Answer:

31. Heat transfer through direct particle contact is called .
Answer:

32. The visible colors of light together form the visible .
Answer:

33. Matter in the state made of charged particles is called .
Answer:

34. A sound wave’s tightly packed regions are called .
Answer:

35. A changing magnetic field can produce electric current by electromagnetic .
Answer:

Section D — Matching

10 marks

36. Match each term to its correct definition.

Column A	Answer	Column B
Voltage		A. Attraction between masses
Faraday		B. Sound above human hearing range
Gravity		C. Electrical potential difference
Conduction		D. Heat transfer by direct contact
Ultrasound		E. Electromagnetic induction

Section E — Short Answer

20 marks

41. What is the difference between a conductor and an insulator?

42. Explain what Oersted’s 1820 observation revealed.

Advanced · 4 pts

43. Why do objects fall at the same rate in a vacuum?

Advanced · 4 pts

44. Distinguish heat from temperature.

Advanced · 4 pts

45. Explain the Doppler effect using an approaching ambulance as an example.

Advanced · 4 pts

Candidate Signature: _____ **Invigilator Signature:** _____

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Marks Obtained: _____ / 100

Checked By: _____

Result / Grade: _____

Date Checked: _____

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Physics Comprehensive Practice Test 2 — Answer Key

Teacher reference: equivalent grammatically correct wording is acceptable for open-ended items when the meaning is preserved.

Section A — Multiple Choice

- | | |
|------|-------|
| 1. C | 9. C |
| 2. B | 10. B |
| 3. C | 11. C |
| 4. C | 12. B |
| 5. C | 13. D |
| 6. C | 14. B |
| 7. C | 15. A |
| 8. C | |

Section B — True / False

- | | |
|-----------|-----------|
| 16. False | 21. True |
| 17. True | 22. False |
| 18. False | 23. False |
| 19. True | 24. True |
| 20. False | 25. False |

Section C — Fill in the Blank

- | | |
|------------|------------------|
| 26. r | 31. conduction |
| 27. south | 32. spectrum |
| 28. 6 | 33. plasma |
| 29. square | 34. compressions |
| 30. kelvin | 35. induction |

Section D — Matching

- | | |
|-----------------|--------------------|
| 36. Voltage — C | 39. Conduction — D |
| 37. Faraday — E | 40. Ultrasound — B |
| 38. Gravity — A | |

Section E — Short Answer (Model Answers)

- 41.** conductor; current; insulator; resist
- 42.** current; magnetic field; wire; compass
- 43.** air resistance; gravity; same acceleration
- 44.** heat; energy transfer; temperature; average kinetic
- 45.** frequency; pitch; compressed; stretched; approach